

Robert Baumgartner

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Software engineer with a focus on C++ and Python. Both a research based profile through a PhD, and an industrial background through multiple positions as a software engineer. Interested in machine learning, fast executing software and software optimizations, algorithms, GPU programming, and anything related. International profile through many stays abroad.

Experience

Software Engineer

Rohde&Schwarz, Munich | Jan. 2025 - today

- Scrum driven software engineering on micro-services based architecture
- Implemented several features in back- and frontend
- C++, C#, gRPC, QT-framework, SQL, unit testing

Backend Software Engineer

cyberTECHNOLOGIES, Eching | June 2020 - Oct. 2020

- Algorithmic speedups via algorithmic or hardware optimizations
- Initial design of architecture for new operating software

Part time Software Engineer

Blickfeld, Munich | Sept. 2019 - Apr. 2020

- Point cloud processing using ROS and OpenCV

Part time Software Engineer

Intel Inc., Munich | Mar. 2018 - Sept. 2018

- Several features and bugfixes in testing tools using Python and C++
- Support in testing setups

Education

PhD Candidate and Postdoctoral Researcher

TU Delft, Delft | Oct. 2020 - Jan. 2025

- Designed and implemented several algorithms for state machine inference in C++
- Profiler driven code optimizations where applicable
- Implemented several neural networks and trained them, such as transformers, auto-encoders, and simple language models

Technical University of Munich

M.Sc. Electrical Engineering | 2017 - 2020

- Focus on embedded software, machine learning, and robotics
- Master thesis in simulation optimization using CUDA and genetic algorithm applications to train a robotic controller

University of Stuttgart

B.Sc. Electrical and Electronic Engineering | 2013 - 2016

- Focus on semiconductor physics and design, and signal processing

Scholarships

Tokyo Institute of Technology Scholarship, 2016

Languages

German (native), English (fluent), Japanese (early conversational)

Teaching

Technical Assistant for course 'Cyber Data Analytics'

TU Delft | Apr. 2021 - Aug. 2024

- Collaborate with professor to support and guide students through practical training via lab assignments.

Tutor for course 'VHDL Laboratory'

TU Munich | Oct. 2018 - April 2019

- Student support for laboratory. Implementation of several encryption algorithms in VHDL and testing on hardware.

Tutor for course 'Linear Algebra'

University of Stuttgart | Oct. 2015 - Feb. 2016

- Collaborate with professor to support and teach students in the basic of linear algebra.

Skills

Software Engineering

C++ - Python - C# - Git - CUDA - Bash - Linux - CMake - QT - gRPC - Algorithmic design - SQL

Machine Learning

PyTorch - TF 2.0 - Data Analysis - Pandas - numpy

Other Skills

Anomaly detection - Basic Japanese - Basic Mandarin

Publications

- Towards robust automata inference from sparse data using transformer models. R. Baumgartner, C. Shibata, S. Verwer (submitted)
- PDFa distillation with error bound guarantees. R. Baumgartner and S. Verwer, 28th Conference on Implementation and Application of Automata, 2024
- PDFa distillation via String Probability Queries. R. Baumgartner and S. Verwer. LearnAUT, 2024
- Database Assisted State Machine Learning. H. Walinga, R. Baumgartner, S. Verwer. LearnAUT, 2024
- Learning State Machines from Data Streams: A Generic Strategy and an Improved Heuristic. R. Baumgartner and S. Verwer. International Conference on Grammatical Inference, 2023
- Sok: Explainable Machine Learning for Computer Security Applications. A. Nadeem, D. Vos, C. Cao, L. Pajola, S. Diek, R. Baumgartner, S. Verwer. 2023 IEEE 8th European Symposium on Security and Privacy (EuroS&P)
- Learning State Machines via Efficient Hashing of Future Traces. R. Baumgartner & S. Verwer. LearnAUT 2022
- Exploring Hybrid CTC/Attention End-to-End Speech Recognition with Gaussian Processes. L. Kürzinger, T. Watzel, L. Li, R. Baumgartner, G. Rigoll. SPECOM 2029